Lesson 3.2 The One-Quarter and Higher Fractions of the 2^k Design

The One-Quarter of the 2^k Design

For a moderately large number of factors, smaller fractions of the 2^k design are frequently useful. Consider a one-quarter fraction of the 2^k design. This design contains 2^{k-2} runs and is usually called a 2^{k-2} fractional factorial.

The 2^{k-2} design may be constructed by first writing down a basic design consisting of the runs associated with a full factorial in $k-2$ factors and then associating the two additional columns with appropriately chosen interactions involving the first $k-2$ factors. Thus, a one-quarter fraction of the 2^k design has two generators. If P and Q represent the generators chosen, then $I = P$ and $I = Q$ are called the **generating relations** for the design. The signs of P and Q (either + or -) determine which one of the one-quarter fractions is produced. All four fractions associated with the choice of generators $\pm P$ and $\pm Q$ are members of the same family. The fraction for which both P and Q are positive is the **principal fraction**.

The complete defining relation for the design consists of all the columns that are equal to the identity column I. These will consist of P, Q, and their generalized interaction PQ; that is, the defining relation is $I = P = Q = PQ$. We call the elements P, Q, and PQ in the defining relation **words**. The aliases of any effect are produced by the multiplication of the column for that effect by each word in the defining relation. Clearly, each effect has three aliases. The experimenter should be careful in choosing the generators so that potentially important effects are not aliased with each other.

As an example, consider the 2^{6-2} design. Suppose we choose $I = ABCE$ and $I = BCDF$ as the design generators. Now the generalized interaction of the generators $ABCE$ and $BCDF$ is $ADEF$; therefore, the complete defining relation for this design is

$$I = ABCE = BCDF = ADEF$$

Consequently, this is a resolution IV design. To find the aliases of any effect (e.g., A), multiply that effect by each word in the defining relation. For A, this produces

$$A = BCE = ABCDF = DEF$$

It is easy to verify that every main effect is aliased by three- and five-factor interactions, whereas two-factor interactions are aliased with each other and with higher order interactions. Thus, when we estimate A, for example, we are really estimating $A+BCE+DEF+ABCDF$.

The complete alias structure of this design is shown in the table below. If three-factor and higher interactions are negligible, this design gives clear estimates of the main effects.

Alias Structure for the 2_{IV}^{6-2} Design with $I = ABCE = BCDF = ADEF$

$A = BCE = DEF = ABCDF$	$AB = CE = ACDF = BDEF$
$B = ACE = CDF = ABDEF$	$AC = BE = ABDF = CDEF$
$C = ABE = BDF = ACDEF$	$AD = EF = BCDE = ABCF$
$D = BCF = AEF = ABCDE$	$AE = BC = DF = ABCDEF$
$E = ABC = ADF = BCDEF$	$AF = DE = BCEF = ABCD$
$F = BCD = ADE = ABCEF$	$BD = CF = ACDE = ABEF$
	$BF = CD = ACEF = ABDE$
$ABD = CDE = ACF = BEF$	
$ACD = BDE = ABF = CEF$	

To construct the design, first write down the basic design, which consists of the 16 runs for a full $2^{6-2} = 2^4$ design in A, B, C, and D. Then the two factors E and F are added by associating their plus and minus levels with the plus and minus signs of the interactions ABC and BCD, respectively. This procedure is shown below.

Construction of the 2_{IV}^{6-2} Design with the Generators $I = ABCE$ and $I = BCDF$

	Basic Design					
Run	***A***	***B***	***C***	***D***	***E = ABC***	***F = BCD***
1	−	−	−	−	−	−
2	+	−	−	−	+	−
3	−	+	−	−	+	+
4	+	+	−	−	−	+
5	−	−	+	−	+	+
6	+	−	+	−	−	+
7	−	+	+	−	−	−
8	+	+	+	−	+	−
9	−	−	−	+	−	+
10	+	−	−	+	+	+
11	−	+	−	+	+	−
12	+	+	−	+	−	−
13	−	−	+	+	+	−
14	+	−	+	+	−	−
15	−	+	+	+	−	+
16	+	+	+	+	+	+

Another way to construct this design is to derive the four blocks of the 2^6 design with $ABCE$ and $BCDF$ confounded and then choose the block with treatment combinations that are positive on $ABCE$ and $BCDF$. This would be a 2^{6-2} fractional factorial with generating relations $I = ABCE$ and $I = BCDF$, and because both generators $ABCE$ and $BCDF$ are positive, this is the principal fraction.

There are, of course, three **alternate fractions** of this particular design. They are the fractions with generating relationships $I = ABCE$ and $I = -BCDF$; $I = -ABCE$ and $I = BCDF$; and $I = -ABCE$ and $I = -BCDF$. These fractions may be easily constructed by the using the table above. For example, if we wish to find the fraction for which $I = ABCE$ and $I = -BCDF$, then in the last column of the table we set $F = -BCD$, and the column of levels for factor F becomes

$$++---++--++++--$$

The complete defining relation for this alternate fraction is $I = ABCE = -BCDF = -ADEF$. Certain signs in the alias structure in the table are now changed; for instance, the aliases of A are $A = BCE = -DEF = -ABCDF$. Thus, the linear combination of the observations [A] actually estimates $A + BCE - DEF - ABCDF$.

Finally, note that the fractional factorial will project into a single replicate of a 2^4 design in any subset of four factors that is not a word in the defining relation. It also collapses to a replicated one-half fraction of a 2^4 in any subset of four factors that is a word in the defining relation. Thus, the design in the previous table becomes two replicates of a 2^{4-1} in the factors $ABCE$, $BCDF$, and $ADEF$, because these are the words in the defining relation. There are 12 other combinations of the six factors, such as $ABCD$, $ABCF$, for which the design projects to a single replicate of the 2^4. This design also collapses to two replicates of a 2^3 in any subset of three of the six factors or four replicates of a 2^2 in any subset of two factors.

An example

Parts manufactured in an injection molding process are showing excessive shrinkage. This is causing problems in assembly operations downstream from the injection molding area. A quality improvement team has decided to use a designed experiment to study the injection molding process so that shrinkage can be reduced. The team decides to investigate six factors—mold temperature (A), screw speed (B), holding time (C), cycle time (D), gate size (E), and holding pressure (F)—each at two levels, with the objective of learning how each factor affects shrinkage and also, something about how the factors interact.

The team decides to use the 16-run two-level fractional factorial design as described earlier. The design is shown again below, along with the observed shrinkage ($\times 10$) for the test part produced at each of the 16 runs in the design.

A 2_{IV}^{6-2} Design for the Injection Molding Experiment in Example 8.4

Run	Basic Design A	B	C	D	E = ABC	F = BCD	Observed Shrinkage (× 10)	
1	−	−	−	−	−	−	6	(1)
2	+	−	−	−	+	−	10	*ae*
3	−	+	−	−	+	+	32	*bef*
4	+	+	−	−	−	+	60	*abf*
5	−	−	+	−	+	+	4	*cef*
6	+	−	+	−	−	+	15	*acf*
7	−	+	+	−	−	−	26	*bc*
8	+	+	+	−	+	−	60	*abce*
9	−	−	−	+	−	+	8	*df*
10	+	−	−	+	+	+	12	*adef*
11	−	+	−	+	+	−	34	*bde*
12	+	+	−	+	−	−	60	*abd*
13	−	−	+	+	+	−	16	*cde*
14	+	−	+	+	−	−	5	*acd*
15	−	+	+	+	−	+	37	*bcdf*
16	+	+	+	+	+	+	52	*abcdef*

The effect estimate for A and is

$$\begin{aligned} A &= \frac{Contrast}{r(2^{6-2-1})} \\ &= \frac{-6+10-32+60-4+15-26+60-8+12-34+60-16+5-37+52}{8} \\ &= 13.875, \end{aligned}$$

while, the effect estimate for AB is

$$\begin{aligned} AB &= \frac{Contrast}{r(2^{6-2-1})} \\ &= \frac{6-10-32+60+4-15-26+60+8-12-34+60+16-5-37+52}{8} \\ &= 11.875, \end{aligned}$$

Remember AB is aliased with CE which means that AB effect is the same as $AB + CE$.

TASK: Calculate the rest of the effect estimates.

We can use the following code chunck to generate the regression and effects estimates.

```
library(tidyverse)
library(FrF2)

#generates the design matrix
set.seed(1234)
design <- as.data.frame(FrF2(nruns = 16,
                             nfactors = 6,
                             generators = c("ABC","BCD")))
design
```

```
##     A  B  C  D  E  F
## 1   1  1 -1  1 -1 -1
## 2   1 -1 -1  1  1  1
## 3   1 -1  1 -1 -1  1
## 4  -1 -1  1 -1  1  1
## 5   1  1  1  1  1  1
## 6  -1 -1 -1  1 -1  1
## 7  -1 -1  1  1  1 -1
## 8   1 -1  1  1 -1 -1
## 9   1  1  1 -1  1 -1
## 10  1  1 -1 -1 -1  1
## 11  1 -1 -1 -1  1 -1
## 12 -1  1 -1  1  1 -1
## 13 -1  1  1  1 -1  1
## 14 -1  1 -1 -1  1  1
## 15 -1  1  1 -1 -1 -1
## 16 -1 -1 -1 -1 -1 -1
```

```
shrinkage <- c(60, 12, 15, 4, 52, 8, 16, 5, 60, 60, 10, 34, 37, 32, 26, 6)

df <- design %>%
  mutate(Y=shrinkage)
df
```

```
##     A  B  C  D  E  F  Y
## 1   1  1 -1  1 -1 -1 60
## 2   1 -1 -1  1  1  1 12
## 3   1 -1  1 -1 -1  1 15
## 4  -1 -1  1 -1  1  1  4
## 5   1  1  1  1  1  1 52
## 6  -1 -1 -1  1 -1  1  8
## 7  -1 -1  1  1  1 -1 16
```

```
## 8   1 -1  1  1 -1 -1  5
## 9   1  1  1 -1  1 -1 60
## 10  1  1 -1 -1 -1  1 60
## 11  1 -1 -1 -1  1 -1 10
## 12 -1  1 -1  1  1 -1 34
## 13 -1  1  1  1 -1  1 37
## 14 -1  1 -1 -1  1  1 32
## 15 -1  1  1 -1 -1 -1 26
## 16 -1 -1 -1 -1 -1 -1  6
```

```r
model <- lm(Y ~ A*B*C*D, data = df)

summary(model)
```

```
##
## Call:
## lm.default(formula = Y ~ A * B * C * D, data = df)
##
## Residuals:
## ALL 16 residuals are 0: no residual degrees of freedom!
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  27.3125        NaN     NaN      NaN
## A1            6.9375        NaN     NaN      NaN
## B1           17.8125        NaN     NaN      NaN
## C1           -0.4375        NaN     NaN      NaN
## D1            0.6875        NaN     NaN      NaN
## A1:B1         5.9375        NaN     NaN      NaN
## A1:C1        -0.8125        NaN     NaN      NaN
## B1:C1        -0.9375        NaN     NaN      NaN
## A1:D1        -2.6875        NaN     NaN      NaN
## B1:D1        -0.0625        NaN     NaN      NaN
## C1:D1        -0.0625        NaN     NaN      NaN
## A1:B1:C1      0.1875        NaN     NaN      NaN
## A1:B1:D1      0.0625        NaN     NaN      NaN
## A1:C1:D1     -2.4375        NaN     NaN      NaN
## B1:C1:D1      0.1875        NaN     NaN      NaN
## A1:B1:C1:D1   0.3125        NaN     NaN      NaN
##
## Residual standard error: NaN on 0 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:    NaN
## F-statistic:   NaN on 15 and 0 DF,  p-value: NA
```

The General 2^{k-p} Fractional Factorial Design

A 2^k fractional factorial design containing 2^{k-p} runs is called a $\frac{1}{2^p}$ fraction of the 2^k design or, more simply, a 2^{k-p} fractional factorial design. These designs require the selection of p independent generators. The defining relation for the design consists of the p generators initially chosen and their $2^p - p - 1$ generalized interactions.

The alias structure may be found by multiplying each effect column by the defining relation. Care should be exercised in choosing the generators so that effects of potential interest are not aliased with each other. Each effect has $2^p - 1$ aliases. For moderately large values of k, we usually assume higher order interactions (say, third- or fourth-order and higher) to be negligible, and this greatly simplifies the alias structure.

It is important to select the p generators for a 2^{k-p} fractional factorial design in such a way that we obtain the best possible alias relationships. A reasonable criterion is to select the generators such that the resulting 2^{k-p} design has the highest possible resolution. To illustrate, consider the 2_{IV}^{6-2} design in the previous example, where we used the generators $E = ABC$ and $F = BCD$, thereby producing a design of resolution IV. This is the maximum resolution design. If we had selected $E = ABC$ and $F = ABCD$, the complete defining relation would have been $I = ABCE = ABCDF = DEF$, and the design would be of resolution III. Clearly, this is an inferior choice because it needlessly sacrifices information about interactions.

The table below presents a selection of 2^{k-p} fractional factorial designs for $k \leq 15$ factors and up to $n \leq 128$ runs. The suggested generators in this table will result in a design of the highest possible resolution.

Number of Factors, k	Fraction	Number of Runs	Design Generators
3	2_{III}^{3-1}	4	$C = \pm AB$
4	2_{IV}^{4-1}	8	$D = \pm ABC$
5	2_{V}^{5-1}	16	$E = \pm ABCD$
	2_{III}^{5-2}	8	$D = \pm AB$
			$E = \pm AC$
6	2_{VI}^{6-1}	32	$F = \pm ABCDE$
	2_{IV}^{6-2}	16	$E = \pm ABC$
			$F = \pm BCD$
	2_{III}^{6-3}	8	$D = \pm AB$
			$E = \pm AC$
			$F = \pm BC$
7	2_{VII}^{7-1}	64	$G = \pm ABCDEF$
	2_{IV}^{7-2}	32	$F = \pm ABCD$
			$G = \pm ABDE$
	2_{IV}^{7-3}	16	$E = \pm ABC$
			$F = \pm BCD$
			$G = \pm ACD$
	2_{III}^{7-4}	8	$D = \pm AB$
			$E = \pm AC$
			$F = \pm BC$
			$G = \pm ABC$
8	2_{V}^{8-2}	64	$G = \pm ABCD$
			$H = \pm ABEF$
	2_{IV}^{8-3}	32	$F = \pm ABC$
			$G = \pm ABD$
			$H = \pm BCDE$
	2_{IV}^{8-4}	16	$E = \pm BCD$
			$F = \pm ACD$
			$G = \pm ABC$
			$H = \pm ABD$
9	2_{VI}^{9-2}	128	$H = \pm ACDFG$
			$J = \pm BCEFG$
	2_{IV}^{9-3}	64	$G = \pm ABCD$
			$H = \pm ACEF$
			$J = \pm CDEF$
	2_{IV}^{9-4}	32	$F = \pm BCDE$
			$G = \pm ACDE$
			$H = \pm ABDE$
			$J = \pm ABCE$
	2_{III}^{9-5}	16	$E = \pm ABC$
			$F = \pm BCD$
			$G = \pm ACD$
			$H = \pm ABD$
			$J = \pm ABCD$
10	2_{V}^{10-3}	128	$H = \pm ABCG$
			$J = \pm BCDE$
			$K = \pm ACDF$
	2_{IV}^{10-4}	64	$G = \pm BCDF$
			$H = \pm ACDF$
			$J = \pm ABDE$
			$K = \pm ABCE$
	2_{IV}^{10-5}	32	$F = \pm ABCD$
			$G = \pm ABCE$
			$H = \pm ABDE$
			$J = \pm ACDE$
			$K = \pm BCDE$
	2_{III}^{10-6}	16	$E = \pm ABC$
			$F = \pm BCD$
			$G = \pm ACD$
			$H = \pm ABD$
			$J = \pm ABCD$
			$K = \pm AB$
11	2_{IV}^{11-5}	64	$G = \pm CDE$
			$H = \pm ABCD$
			$J = \pm ABF$
			$K = \pm BDEF$
			$L = \pm ADEF$
	2_{IV}^{11-6}	32	$F = \pm ABC$
			$G = \pm BCD$
			$H = \pm CDE$
			$J = \pm ACD$
			$K = \pm ADE$
			$L = \pm BDE$
	2_{III}^{11-7}	16	$E = \pm ABC$
			$F = \pm BCD$
			$G = \pm ACD$
			$H = \pm ABD$
			$J = \pm ABCD$
			$K = \pm AB$
			$L = \pm AC$
12	2_{III}^{12-8}	16	$E = \pm ABC$
			$F = \pm ABD$
			$G = \pm ACD$
			$H = \pm BCD$
			$J = \pm ABCD$
			$K = \pm AB$
			$L = \pm AC$
			$M = \pm AD$
13	2_{III}^{13-9}	16	$E = \pm ABC$
			$F = \pm ABD$
			$G = \pm ACD$
			$H = \pm BCD$
			$J = \pm ABCD$
			$K = \pm AB$
			$L = \pm AC$
			$M = \pm AD$
			$N = \pm BC$
14	2_{III}^{14-10}	16	$E = \pm ABC$
			$F = \pm ABD$
			$G = \pm ACD$
			$H = \pm BCD$
			$J = \pm ABCD$
			$K = \pm AB$
			$L = \pm AC$
			$M = \pm AD$
			$N = \pm BC$
			$O = \pm BD$
15	2_{III}^{15-11}	16	$E = \pm ABC$
			$F = \pm ABD$
			$G = \pm ACD$
			$H = \pm BCD$
			$J = \pm ABCD$
			$K = \pm AB$
			$L = \pm AC$
			$M = \pm AD$
			$N = \pm BC$
			$O = \pm BD$
			$P = \pm CD$

In summary a 2^{k-p} design

1. requires p independent generators,
2. the defining relation consists of p generators and their $2^p - p - 1$ generalized interactions, and
3. each effect is aliased with $2^p - 1$ aliases.

An example [A classroom exercise!]

Consider a design for analyzing 7 factors in only 8 runs (2^{7-4}). Use the following as generators: ABD, ACE, BCF, and $ABCG$.

1. Construct the design. Indicate the treatment runs.
2. Give the complete defining relations.
3. Give the aliases of B.
4. What is the resolution of the design?
5. Suppose the 3-way and higher-order interactions are negligible, what are the alias structure for all 7 factors?